# DGA Classifier from Domain Names: Family-First or Direct Binary?

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## 1 Problem and Motivation

Malware uses Domain Generation Algorithms (DGAs) to generate many domains from a shared seed, so infected hosts can reach the attacker without a hard-coded address that defenders could block [7]. Real detection also uses network signals such as the NXDOMAIN rate and query periodicity [1]; this project focuses on detecting DGAs from the domain name alone. The difficulty is that "DGA" is not one pattern: some families produce random strings (`gqspwpsurdun.ru`, CryptoLocker) or hex hashes (`2c3f1e3326cca35b757ec704b2413a30.org`, Bamital), while others join dictionary words (`kneemanualgirl.com`, `whetherspread.net`) and look much closer to legitimate names.

**Question.** When the final decision is DGA vs. legitimate, should a classifier first learn the individual DGA families (each family + legitimate, with the prediction collapsed to DGA/legitimate) or learn DGA vs. legitimate directly?

**Why it matters.** The answer is for teams that build name-based DGA filters, for example in DNS resolvers or security products, where a model must judge a single query before any connection is made. If family-first training wins, family labels are worth collecting even when only a binary decision is needed, and simpler models may become accurate enough to deploy. If it does not, the simpler binary setup is enough.

**Goal.** For each model we evaluate, measure the difference in macro-F1 on the DGA/legitimate decision between family-first and direct binary training, on the same test set and averaged over 5 random seeds. We count family-first as better when its mean macro-F1 is higher by more than the standard deviation across seeds.

## 2 Impact

**If it succeeds.** Teams that build name-based DGA filters get a clear rule for how to train them: if family-first wins, it is worth collecting and maintaining family labels even when only a DGA/legitimate verdict is shipped; if both setups tie, the cheaper binary labels are enough. The per-family results also show which kinds of DGA (random-string or dictionary-based) a name-only detector can and cannot catch, which tells analysts where to expect blind spots.

**If it fails.** If neither setup separates some families from legitimate domains, in particular dictionary-based ones, that is evidence that the name alone does not carry enough signal for them. Detection would then need network traffic features, such as the NXDOMAIN rate, the periodicity of queries, and how many distinct, previously unseen domains a single host queries [1]. Knowing which families need these signals is itself a useful result.

**Risks of using the result.**

- *Errors do not cost the same.* A missed DGA domain (false negative) leaves an infected host talking to its attacker. A legitimate domain flagged as DGA (false positive) blocks a real service and adds to alert fatigue. Since almost all real DNS traffic is legitimate, even a small false-positive rate produces many false alarms.
- *Bias in the data.* Legitimate examples come from a popularity ranking (Tranco [5]), which over-represents popular sites. Legitimate but unusual names, such as non-English, short or machine-generated CDN hostnames, may be flagged unfairly. Families with fewer than 300 known domains are left out of our data, so our results say nothing about them.
- *Limited validity.* We test only families seen in training. Attackers can change their algorithms, so results do not guarantee detection of new or adversarial DGAs.

# 3 Related Work

# 4 Data

**Sources.** We combine two DGA sources and one legitimate source into one table that gives, for each domain, its label (DGA/legitimate) and its family.

- *DGA (main):* "Legitimate domains and DGA categorized morphologically and by families" by Gregório and Cansian, Mendeley Data v1, 2025, `https://data.mendeley.com/datasets/nhvyvytn2h/1`, license CC BY 4.0 [3]. We use its 3,090,661 DGA domains, collected from the DGArchive feed [7] (28 Sep–28 Oct 2025) and labelled with 160 families.
- *DGA (extra):* Kaggle "DGA dataset" by C. Givre (v1, 2021, `https://www.kaggle.com/datasets/gtkcyber/dga-dataset`, license not stated, used for academic research only) [2]. We use its 80,000 DGA domains (7 family labels).
- *Legitimate:* the Tranco top 1M list [5] (list `64VVX`, `https://tranco-list.eu/list/64VVX`), which ranks pay-level domains by combining the rankings of five providers (Chrome UX Report, Farsight, Majestic, Cloudflare Radar, Cisco Umbrella) over 17 Sep–16 Oct 2026.

The legitimate parts of the two DGA datasets (Majestic and Alexa) are not used, so all legitimate domains come from one documented ranking.

**Size, features and target.** The dataset we use has 439,974 domains: 219,987 legitimate and 219,987 DGA from 102 families, with 300 to 3,000 domains per family (54 families have 3,000), so the largest family is at most 10 times the smallest (Fig. 1, Table 1). The only model input is the domain string (4 to 93 characters; a–z, 0–9, '-' and '.'); any numerical features are derived from it. The targets are the binary label (DGA/legitimate) and the family (`legit` or one of the 102 DGA families). Data were loaded with pandas [6] and plotted with Matplotlib [4].

Table 1: Statistics per class. Entropy is the Shannon entropy (bits) of the characters before the TLD; digit ratio is the share of digits in the same part.

| Class | Domains | Families | Mean length | Entropy | Digit ratio | TLDs | Top TLD |
|---|---|---|---|---|---|---|---|
| Legitimate | 219,987 | – | 13.8 | 2.80 | 0.038 | 792 | .com (42%) |
| DGA | 219,987 | 102 | 19.1 | 3.18 | 0.148 | 235 | .com (24%) |

**Missing values, noise and biases.** There are no empty cells. We found:

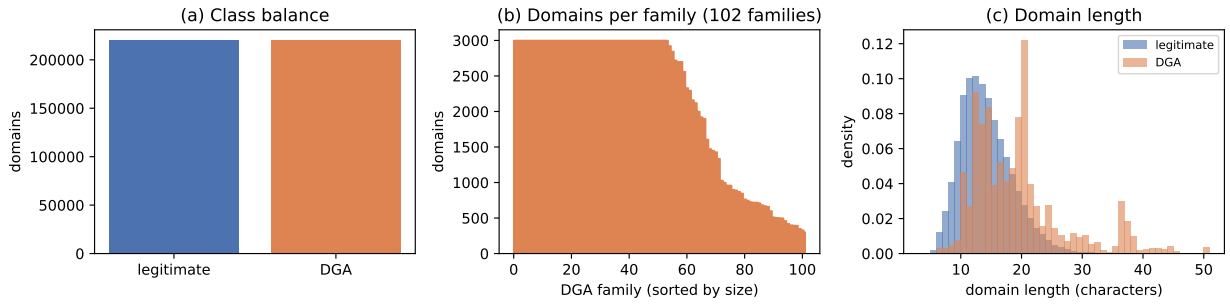



Figure 1: (a) Number of legitimate and DGA domains; (b) number of domains in each of the 102 DGA families; (c) domain length of legitimate and DGA domains.

- *Label conflicts:* 330 domains are in both Tranco and a DGA list. Some are short DGA names that coincide with real sites (e.g. `qsnatch` generates 3–4 letter names); others are clearly generated (e.g. `pywolwnvd.biz`, family `m0yv`) and likely entered Tranco through the DNS traffic of infected machines. We drop all 330.
- *Noise:* 141 duplicate DGA domains, 6 Tranco wildcard entries (e.g. `_wildcard_.ph`) and one domain containing a line break. The Kaggle `domain` column is truncated at hyphens for some rows, so we use its `host` column.
- *Family names:* no DGA domain appears in both DGA sources (they are years apart), so Kaggle labels were matched by domain shape: `newgoz` and `gameoverdga` → `gameover`, `goz` → the GameOver P2P family, `nivdort` kept as a new family.
- *Biases:* Tranco favours popular sites; each DGA family uses few TLDs (median 3.5, and 23 families use a single one, vs. 792 for legitimate domains), so the TLD alone may act as a shortcut; DGA data cover two short periods (around 2016 and 2025).

**Preprocessing and balancing.** We lower-case and validate domains and drop conflicts and duplicates, which leaves 999,664 legitimate and 3,170,189 DGA domains from 161 families of very different sizes (2 to 519,280 domains). We then balance: we keep the families with at least 300 domains (59 smaller families are left out), sample at most 3,000 domains per family, and draw the same number of legitimate domains uniformly at random from the cleaned Tranco list. The remaining 10× family imbalance is handled with class weights in the loss.

## References

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